



PHILOSOPHY OF PHYSICS

Evaluating String Theory Through the Lens of Scientific Realism



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Abstract

String theory has become immensely popular in the last few decades. It is viewed as a strong candidate for unifying quantum mechanics and general relativity, but it has also been criticised. Most of the debate is centred around scientific realism. In this project, we discuss some of the arguments in favour of and against scientific realism and then apply them to string theory.

1 Introduction

The 20th century was a century of significant breakthroughs in physics. Two of these are the developments of quantum mechanics (QM) and general relativity (GR); both theories are now

foundational to modern physics. The two revolutionized our perception of the physical world, but in different ways. QM describes the world at very small scales, like the scale of subatomic particles. On the other hand, GR describes the world at large scales, e.g. black holes and entire galaxies. Naturally, the question arises if it is possible to unify the two.

Physicists have tackled this question for quite some time. In the past few decades, a theory called *string theory* has become very popular. In this theory, the foundational constituents of the universe are described as one-dimensional strings instead of zero-dimensional point particles as in quantum field theory. As of now, string theory seems to be the most promising unifying theory [1].

Despite this, there has been some criticism as well. These criticisms come especially from the community of philosophers of physics/science. A lot of the debate is centred around an idea called *scientific realism*, which states that scientific theories give a true picture of what the world looks like [2].

In this project, we start in *Section 2* by introducing the main ideas of string theory. In *Section 3*, we set up the philosophical conceptual framework by introducing the idea behind scientific realism and discussing one of the most important arguments in favour of and against scientific realism. In *Section 4*, we discuss some of the arguments for and against string theory in the context of the arguments presented in the previous section. Our findings will be discussed in the conclusion.

2 String Theory Overview

In this section, before discussing the limitations from the perspectives of the philosophy of physics and science, we introduce string theory as a physical theory. We provide a concise summary of its essential concepts and motivations. This overview aims to establish a foundational understanding of why the theory is highly regarded in the scientific community, despite raising significant philosophical concerns.

2.1 Core Ideas & Unification of QM and GR

String theory proposes that the fundamental constituents of the universe are not zero-dimensional point particles, as traditionally conceived in quantum field theory, but one-dimensional vibrating *strings*. Specifically, they can be *open strings* (with two distinct endpoints) or *closed strings* (forming closed loops). Different vibrational modes of these strings correspond to different fundamental particles, including force carriers (e.g. photons) and massive particles (e.g. gravitons).



Figure 1: closed string and open string [3].

Strings exist in a higher-dimensional spacetime, beyond the familiar four (three spatial and one temporal) dimensions.¹ The extra dimensions are hypothesised to be compactified or curled up into extremely small geometrical structures comparable to the Planck length ($\sim 10^{-35}$ m), such as *Calabi-Yau manifolds*, which allow them to exist without contradicting observations of the large-scale universe [1].

Unlike GR, which breaks down at the Planck scale due to the emergence of *singularities*,² string theory eliminates these singularities by using one-dimensional strings [4]. The finite size of strings spreads interactions over a small region, preventing the infinite values that arise in point-like interactions [2].

¹*Bosonic string theory* suggests 26 dimensions (25 spatial and 1 temporal), while *superstring theory* requires 10 dimensions (9 spatial and 1 temporal), and *M-theory* unifies these frameworks in an 11-dimensional spacetime (10 spatial and 1 temporal).

²Singularities are the points where physical quantities like curvature become infinite, e.g. black hole singularities or the Big Bang.

D -branes play a crucial role in string theory, arising naturally when considering strings and their interactions. While strings are inherently one-dimensional objects, D -branes expand the theory by incorporating higher-dimensional surfaces that strings can interact with or terminate on.

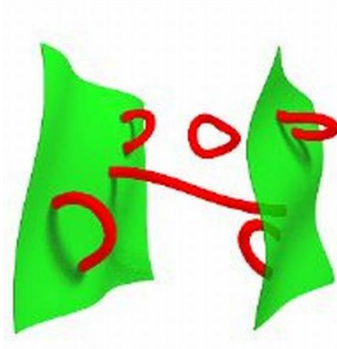


Figure 2: D -brane (created by Steuard Jensen).

Definition 2.1 (Dp -Brane, [3]). A Dp -brane is a p -dimensional object embedded in a D -dimensional spacetime. D represents the endpoints of the open string, which satisfy Dirichlet boundary conditions.

- $p = 0$: *point particle*; traces a *worldline*.
- $p = 1$: *string*; traces a *worldsheet*.
- $p = 2$: *membrane*; sweeps a *worldvolume*.

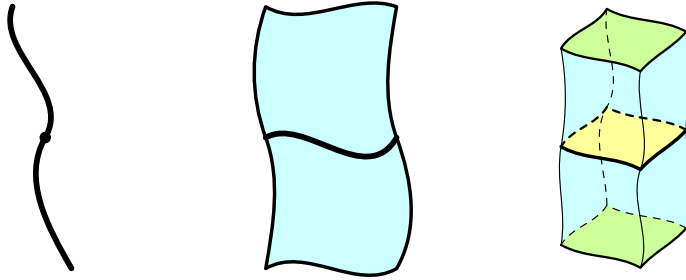


Figure 3: worldline, worldsheet, and worldvolume [3].

2.2 Unification of Gauge Interactions and Gravity

String theory unifies *gauge interactions* (e.g. electromagnetism, weak, and strong forces) with *gravity* by providing a consistent framework in which both phenomena emerge from the vibrational modes of strings.

Theorem 2.2 (The Polyakov Action, [5]). *The Polyakov action is expressed as:*

$$S = -\frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{-g} g^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}, \quad g \equiv \det g,$$

where α' is related to the string tension; X^μ are the spacetime coordinates; $g_{\alpha\beta}$ is the worldsheet metric and $\eta_{\mu\nu}$ is the flat spacetime metric.

The *Polyakov action* is the foundation for analysing string dynamics. By applying different boundary conditions to the general solution of wave equations, one can generate different types of particles [5].

- (i) Applying the appropriate boundary conditions, the general solution for the open string coordinates can be expanded as:

$$X^\mu(\tau, \sigma) = x^\mu + 2\alpha' p^\mu \tau + i\sqrt{2\alpha'} \sum_{n \neq 0} \frac{1}{n} \alpha_n^\mu e^{-in\tau} \cos(n\sigma),$$

where x^μ represents the initial position, p^μ represents the momentum, and α_n^μ is the mode expansion coefficients. Using the Virasoro constraints,³ the mass spectrum is given by:

$$M^2 = \frac{1}{\alpha'}(N - 1),$$

where $N = \sum_{n>0} n \alpha_{-n}^\mu \alpha_n^\mu$ represents the excitation number operator. For $N = 1$, $M^2 = 0$, corresponding to a massless spin-1 gauge boson (e.g. a photon).

- (ii) Closed strings obey periodic boundary conditions:

$$X^\mu(\tau, \sigma) = X^\mu(\tau, \sigma + 2\pi).$$

The mode expansion for closed string coordinates can be written as:

$$X^\mu(\tau, \sigma) = X^\mu + 2\alpha' p^\mu \tau + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{1}{n} \left(\alpha_n^\mu e^{-in(\tau-\sigma)} + \tilde{\alpha}_n^\mu e^{-in(\tau+\sigma)} \right),$$

where α_n^μ and $\tilde{\alpha}_n^\mu$ are mode coefficients for left- and right-moving waves on the string. The Virasoro constraints lead to the mass spectrum

$$M^2 = \frac{4}{\alpha'}(N - 1),$$

where $N = \sum_{n>0} \alpha_{-n}^\mu \alpha_n^\mu = \tilde{N}$. For $N = 1$, $M^2 = 0$, corresponding to a massless spin-2 particle, interpreted as the graviton.

Therefore, string theory provides a natural explanation for the emergence of gauge bosons and gravitons from the same theoretical framework: open strings give rise to gauge bosons, and closed strings predict the graviton.

In conclusion, string theory is one of the most ambitious frameworks in physics, aiming to unify the four fundamental forces of nature: electromagnetism, weak interaction, strong interaction, and gravity. It naturally incorporates both QM and GR, offering a potential ‘theory of everything’. Additionally, string theory predicts the existence of supersymmetric particles and has deep connections with mathematics, including geometry and topology [1].

3 Philosophical Conceptual Framework

After having introduced string theory, we now turn our attention to the philosophical concepts relevant to our project. In particular, we discuss the notion of scientific realism and some of the arguments for and against it.

³The Virasoro constraints arise from reparameterisation invariance and conformal invariance on the string worldsheet, and it aims to ensure that the energy-momentum tensor of the string vanishes [5].

3.1 Scientific Realism

Over the years, there have been different views on the topics of what the aims of science should be and how we should interpret scientific theories. The two main philosophical movements regarding these topics are *logical empiricism*⁴ and *scientific realism*. We only discuss the latter in this project.

As the name suggests, scientific realism argues that we should be realists about our scientific theories. To be more precise, we have the following definition of scientific realism.

Definition 3.1 (Scientific Realism, [6]). Science aims to give us, in its theories, a literally true story of what the world is like; and acceptance of a scientific theory involves the belief that it is true.

This idea is not universally accepted in the scientific community. Several rival epistemologies of science have contested it. All these are forms of scientific anti-realism. Some notable (contemporary) anti-realists include Larry Laudan and Bas van Fraassen. On the other hand, contemporary realists include Richard Boyd and Sebastian de Haro.

3.2 The No Miracles Argument

According to the Stanford Encyclopedia of Philosophy, the no miracles argument (NMA) is ‘the most powerful intuition motivating realism’ [7]. The idea is based on Putnam, who argued that realism ‘is the only philosophy that does not make the success of science a miracle’ [7]. In short, the argument can be put in the following logical form:

P1: Our best theories are extraordinarily successful.

P2: Truth is a necessary requirement for success.

C: Our best theories are true.

Since this is a deductive argument, the conclusion follows necessarily from the premises. Thus, to refute the conclusion, one should respond to (at least one of) the premises. *Premise 1* has been widely accepted, but *Premise 2* has been contested. According to some philosophers, there is no relation between truth and success. See for example [6]. For the purpose of this project, we will not go into the various refutations of this argument, and of the arguments to come.

3.3 Constructive Empiricism

Constructive empiricism, as articulated by Bas van Fraassen, represents a significant stance in contemporary philosophy of science. Emerging as both a continuation and critique of the logical positivist tradition, van Fraassen posits that the primary goal of science is not to deliver a true representation of the world in its entirety, but rather to construct theories that are *empirically adequate* [8].

Definition 3.2 (Empirically Adequate, [9]). A theory is *empirically adequate* if and only if what it says about the observable phenomena in the world is true. Formally, a scientific theory T is empirically adequate if:

$$\forall O \in \mathcal{O}, \quad \mathbb{P}(O \mid T) = 1,$$

where O represents an observable phenomenon, and $\mathcal{O}$ is the set of all observable entities.

Empirical adequacy focuses on the theory’s ability to align with observational data, rather than its capacity to describe unobservable reality. This distinction necessitates a division between observable and unobservable entities [9].

- Observable entities: those that can be directly perceived by human senses or with instruments (e.g. telescopes, microscopes).

⁴For a detailed discussion on logical empiricism, see [6].



- Unobservable entities: those that cannot be directly perceived, even with instruments, such as strings, D -branes, or Calabi-Yau manifolds.

Van Fraassen emphasises that adherence to empirical adequacy, rather than an ontological commitment to unobservable entities, suffices for the success of scientific theories. He argues that scientific practice should focus on constructing models that accurately reflect observable phenomena, without necessarily asserting the literal existence of theoretical entities [9]. This perspective can be formalised as follows:

P1: If a scientific theory is empirically adequate,
it fulfills the primary aim of science.

P2: Unobservable entities introduce unnecessary metaphysical assumptions.

C: Ontological commitment to unobservable entities is not required
for a scientific theory to be justified.

3.4 The Pessimistic Meta-Induction Argument

The *pessimistic meta-induction argument* (PMA) draws from historical observations showing that many scientific theories once regarded as true and widely accepted have ultimately been discarded as false [6].⁵ It can be formalised as follows.

P1: Many scientific theories once considered true have ultimately been abandoned as false.

P2: No reason to believe our current best scientific theories will avoid the same fate.

C: It is likely that our current best scientific theories are also false.

Formally, the probability of our currently accepted scientific theories, T_c , being true, given the historical record of falsified theories, approaches zero:

$$\mathbb{P}(T_c \mid T_1, T_2, \dots \text{ were false}) \rightarrow 0.$$

While specific theoretical interpretations may change, core structural components of successful theories often persist across theory changes. It reminds us that scientific theories are provisional and subject to revision, even as they may capture truths about the natural world [10].

4 Scientific Realism in String Theory

4.1 Arguments for String Theory

We now discuss three reasons why string theorists have trust in their theory. All of these arguments are linked back to the NMA discussed in *Section 3.2*. This section is mostly based on the book by Richard Dawid, see [1].

1. The first argument is called ‘the plain no alternatives argument (NAA)’[1]. String theory is a strong candidate for a unified theory of elementary particle interactions and gravity. In fact, string theorists believe that it is the *only* candidate that provides a complete unified description of all the forces in nature. The reason such a unification is so difficult is because gravity is non-renormalizable within the conventional framework. String theory solves this problem and it seems like this is the only theory that can solve this issue.

⁵For example, the phlogiston theory was replaced by Lavoisier’s oxygen theory, and the caloric theory was abandoned after the advent of thermodynamics. Also, many mathematically sophisticated theories have ultimately been discarded.

We can use the idea of NMA: it would be a miracle if string theory would not be a true description of what the world looks like, given that string theory provides a unification of two of the best physical theories we have and that string theory is the *only* theory capable of this.

The latter statement is quite bold and can be argued against: the fact that we do not have another such unification theory does not mean that there exists none. The scientists of today may lack the creativity; it could very well be that, somewhere in the future, we come up with such a theory.

2. The idea of the second argument is that string theory is viewed as a natural continuation of the highly successful research program for particle physics. By applying NMA to the standard model, we can conclude that the entities postulated in the standard model are real. Thus, since string theory is a natural continuation of the standard model, string theory is also a true theory. Furthermore, the theory of gravity has been successful and this successful theory is incorporated in string theory. This gives reason to believe in the latter.

This argument has the same problem as the previous one: there could be another theory that encompasses QM and GR. This is called underdetermination; the theory choice is underdetermined by the data/the knowledge that we have. Why then should we believe string theory is the true one, among all these other potential theories? If one wants to believe this is the case, clearly there is more needed than the two arguments presented (notwithstanding, they do help). This brings us to the final argument.

3. This argument is called ‘the argument of unexpected explanatory coherence (UEA)’[1]. It is generally believed that a good scientific theory is one that predicts novel facts. For example, GR precisely predicted the extra precession in Mercury’s orbit and particle physics predicted the existence of a new particle known as the Higgs. Just as in these examples, the novel facts usually include a prediction that is later confirmed by experiments. However, there is another possibility. Sometimes, introducing a new theoretical principle leads to a more coherent theoretical framework after the theoretical implications of the new principle have been understood more clearly. This leads to unforeseen deeper explanations of theoretical concepts or connections between facts that were once thought to have no connection at all. Examples of these in string theory include

- the prediction of the existence of gravity;
- the concept of supersymmetry is put into a coherent perspective;
- string theory offers a foundational framework for understanding black hole entropy, which is an issue that comes up when GR goes into the quantum world;
- it offers new tools and perspectives in mathematics.

There are many more profound theoretical implications of string theory. Scientific realists would again argue that this would be a miracle if string theory is not a literally true description of the world.

4.2 Arguments against String Theory

While the previous section outlined several arguments supporting string theory, these are not without challenges. In this section, we examine the limitations and objections to string theory.

4.2.1 PMI against Historical Precedents

NAA does not guarantee the correctness of string theory. *Kaluza-Klein theory* was a precursor to modern higher-dimensional theories.⁶ Despite its elegance and theoretical appeal, it failed to

⁶Kaluza-Klein theory unifies electromagnetism and gravity by introducing an extra spatial dimension, showing that electromagnetism can emerge naturally from higher-dimensional GR. It serves as a conceptual precursor to string theory [11].

produce testable predictions and was eventually sidelined [11]. The lack of current alternatives may reflect epistemic limitations rather than the unique correctness of string theory.

Many discarded theories also displayed UEA. For example, the phlogiston theory explains combustion, and the caloric theory explains heat transfer efficiently. Therefore, coherence within a theoretical framework does not necessarily imply truth, and such a historical pattern weakens the inference from internal coherence to truth. PMI further reinforces this scepticism. It highlights the historical record of scientific theories, many of which were initially celebrated for their explanatory power but later replaced. It suggests that current non-empirical arguments in favour of string theory may be equally vulnerable to the same historical trajectory. Consequently, inductive reasoning from past successes to the correctness of current theories remains tenuous.

4.2.2 Unobservable Entities and Empirical Inaccessibility

A central challenge to string theory lies in its reliance on unobservable entities. String theory introduces a range of theoretical constructs, including strings, D -branes, and higher-dimensional spaces. These entities are not merely difficult to observe; they lie fundamentally beyond our current technological capability. While these entities are theoretically rich and mathematically consistent, their empirical inaccessibility remains a significant challenge.

Despite our four-dimensional observable universe, string theorists propose that additional dimensions are compactified into Calabi-Yau manifolds. This compactification bridges the high-dimensional structure with the four-dimensional spacetime we observe, giving rise to effective field theories at low energies. However, string theory allows for many possible compactifications and introduces an immense *landscape problem* [1].⁷ Each possible compactification leads to a different low-energy effective theory, and as such, string theory does not yield unique, testable predictions for physical observables.

For instance, while the Higgs boson was discovered by CERN at the Large Hadron Collider (LHC) in 2012 at an energy of around 125 GeV, string theory does not predict this value uniquely due to its vast landscape of solutions [1]. Some more general predictions are more difficult to confirm. String theory predicts that particles arise from the vibrational modes of strings, with characteristic energy scales corresponding to the Planck energy:

$$E_{\text{Planck}} = \sqrt{\frac{\hbar c^5}{G}} \approx 1.22 \times 10^{19} \text{ GeV}.$$

Reaching such energy scales is far beyond the capabilities of current or foreseeable experimental facilities.

The underdetermination thesis, attributed to Pierre Duhem and W. V. O. Quine, asserts that for any given set of empirical data, multiple theories can potentially explain it [6]. Even if string theory were empirically testable in the future, underdetermination implies that different compactifications might yield identical or nearly identical observable predictions. This would make it impossible to empirically distinguish between them, further complicating the issue of theory choice.

Moreover, string theory suggests predictions of observable effects from higher-dimensional phenomena,⁸ yet gravitational and high-energy experiments have found no evidence so far. For string theory to progress beyond its current status, either new experimental avenues must be discovered or its theoretical framework must yield testable, unique predictions within the scope of accessible physics.

5 Conclusion

We have discussed the various arguments in favour of and against string theory, in the context of scientific realism. Scientists can accept it as a useful mathematical framework for exploring theoretical physics. But until it produces verifiable empirical predictions, belief in its unobservable entities remains unjustified.

⁷The estimated number of possible ground states exceeds 10^{500} . *String landscape* dealing with ground states is an important task for string theory today (Susskind, 2003).

⁸Such as deviations in gravitational forces at sub-millimetre scales or the detection of Kaluza-Klein modes [1].

String theory’s mathematical elegance, predictive richness, and ability to address long-standing problems suggest it may represent a significant step towards a deeper understanding of nature. However, the theory faces substantial philosophical challenges. Its reliance on unobservable entities, such as extra dimensions and *D*-branes, places it in tension with the empiricist demands of constructive empiricism. The landscape problem and the lack of unique predictions further exacerbate these concerns, making it susceptible to PMA.

String theory aligns best with *structural realism*, where belief is placed not in the physical existence of strings or *D*-branes, but in the mathematical and structural relationships that the theory describes. Dualities in string theory (e.g. *T*-duality and *S*-duality) suggest that our understanding of physical ontology might be secondary to the mathematical consistency and structural coherence of the theory [2].

In conclusion, string theory represents a critical case study for both scientific realism and the philosophy of physics. While its empirical shortcomings challenge traditional notions of scientific realism, its mathematical structure and theoretical successes invite us to reconsider the relationship between theory, observation, and reality. Whether string theory will eventually transition from a speculative framework to an empirically validated theory remains an open question, but its role in advancing our conceptual understanding of the universe is undeniable.

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